Introduction to machine learning -10-

Probabilistic methods

Maximum likelihood

Example: D : coin flip 45 tails 55 heads

$D = (x^{(1)} \dots x^{(100)})$ independent identically distr. rand. variables

Likelihood (probability), with respect to a model and its parameters θ

$\mathcal{L}(\theta)$

Example θ : probability of heads

$\mathcal{L}(\theta) = \theta^{N_H}(1-\theta)^{N_T}$ (sequence and not just counts)

$\mathcal{L}(\theta)$ is VERY small

$\mathcal{L}(0,5) = 0,5^{100} \approx 10^{-30}$

log-likelihood

logarithm: strictly monotonic functio
(does not change positio
of maximum)

$$\ell(\theta) = \log \mathcal{L}(\theta) = N_H \log \theta + N_T \log(1-\theta)$$

$\ell(\theta)$ is maximal if derivative is 0

$$0 = \frac{d\ell}{d\theta} = \left(N_H \log \theta + N_T \log(1-\theta) \right) =$$

$$= \frac{N_H}{\theta} - \frac{N_T}{1-\theta}$$

$$\theta_{ML} = \frac{N_H}{N_H + N_T}$$

Example 2: Gaussian distribution

$$p(x^{(i)}) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(- \frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right)$$

μ : mean , σ : variance

$$0 = \ell(\mu, \theta) = \log \prod_{i=1}^{N} \left[\frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(- \frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right) \right] =$$

$$= \sum_{i=1}^{N} \log \left[\frac{1}{\sqrt{2\pi}\,\sigma} \exp(\dots) \right] =$$

$$= \sum_{i=1}^{N} \left[-\frac{1}{2} \log(2\pi) - \log \sigma - \frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right]$$

$$0 = \frac{\partial \ell}{\partial \mu} = \frac{1}{2\sigma^2} \sum_{i=1}^{N} \frac{\partial}{\partial \mu} \left(x^{(i)} - \mu \right)^2 =$$

$$= \frac{-1}{\sigma^2} \sum_{i=1}^{N} \left(x^{(i)} - \mu \right) \Rightarrow \mu = \frac{1}{N} \sum_{i=1}^{N} x^{(i)}$$

$$0 = \frac{\partial \ell}{\partial \sigma} = -\sum_{i=1}^{N} \frac{\partial}{\partial \sigma} \left(\log \sigma + \frac{1}{2\sigma^2} \left(x^{(i)} - \mu \right)^2 \right) =$$

$$= \sum_{i=1} -\frac{1}{\sigma} + \frac{1}{\sigma^3} \left(x^{(i)} - \mu \right)^2$$

$$\sigma_{ML} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(x^{(i)} - \mu \right)^2}$$

Likelihood $p(D|\theta)$ probability of the observations given the parameters

Bayesian $p(\theta|D)$ probability of the parameters given the observation (posterior distribution)

$$p(\theta|D) = \frac{p(\theta)\, p(D|\theta)}{\int p(\theta')\, p(D|\theta')\, d\theta'}$$

Dropping the normalization:

$$p(\theta|D) \propto p(\theta)\, p(D|\theta)$$

(Future observations

$$p(D'|D) = \int p(\theta|D)\, p(D'|\theta)\, d\theta$$

)

$p(\theta)$ prior distribution, our prior belief about data prior to measurement

How to choose $p(\theta)$?

$p(\theta) = 1 \rightarrow$ uninformative anything is possible

but we know that $\theta \sim 0.5$

beta distribution

$$p(\theta; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\,\Gamma(b)}\, \theta^{a-1}(1-\theta)^{b-1}$$

Generalized binomial dist.

$$p(\theta; a,b) \propto \theta^{a-1}(1-\theta)^{b-1}$$

($a=b=1$ uniform)

$$p(\theta, D) \propto p(\theta)\, p(D|\theta) =$$

$$= \left[\theta^{a-1}(1-\theta)^{b-1}\right]\left[\theta^{N_H}(1-\theta)^{N_T}\right] =$$

$$= \theta^{a-1+N_H}(1-\theta)^{b-1+N_T}$$

Expectation value

next flip is head

$$\theta_{pred} = P_r(x' = H \mid D) = \int p(\theta|D)\, P_r(x'=H|\theta)\, d\theta =$$

$$= \int \text{Beta}(\theta, N_H+a, N_T+b)\cdot\theta\, d\theta =$$

$$= E_{\text{Beta}(\ldots)}[\theta] = \frac{N_H + a}{N_H + N_T + a + b}$$

posterior predictive distribution

Maximum a-posteriori (MAP) approx

$$\hat{\theta}_{MAP} = \arg\max_{\theta} p(\theta|D)$$

→ Maximum likelihood

It looks for the mode of the "beta function"

Mode of the beta function $\frac{\alpha-1}{\alpha+\beta-2}$

$$\theta_{MAP} = \frac{N_H + a - 1}{N_H + N_T + a + b - 2}$$

$a = b = 2$

Example: $N_H = 2, N_T = 0$ $\quad N_H = 55 \quad N_T = 45$

$$\hat{\theta}_{ML} = \frac{N_H}{N_H + N_T} \qquad 1 \qquad \frac{55}{100}$$

$$\hat{\theta}_{Bayes} = \frac{N_H + a}{N_H + N_T + a + b} \qquad \frac{4}{6} \qquad \frac{57}{104}$$

$$\hat{\theta}_{MAP} = \frac{N_H + a - 1}{N_H + N_T + a + b - 2} \qquad \frac{3}{4} \qquad \frac{56}{102}$$